An Earthquake Alert system using Internet of Things

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***Abstract*—This project introduces an innovative earthquake detection system leveraging accelerometers for seismic sensing. By incorporating accelerometers from widely available devices like smartphones into a distributed sensor network, the system aims to provide a cost-effective and scalable solution for enhancing seismic monitoring capabilities. Real-time data processing, supported by advanced signal processing and machine learning algorithms, enables the system to differentiate between normal vibrations and seismic activities. Upon detecting anomalies, the system generates alerts to facilitate swift response and early warning mechanisms. Noteworthy features of this system include its affordability, easy deployment, and suitability for implementation in diverse environments, both urban and remote. The project contributes to advancing earthquake detection technologies, thereby bolstering preparedness and resilience against seismic events**

Keywords*— *Accelerometer, Arduino, Earthquake, Seismograph, Vibration

I. Introduction

Humanity's greatest apprehension revolves around natural calamities, and although they occur infrequently, their potential for widespread devastation poses a significant threat to both property and the delicate balance of ecosystems. Among these natural disasters, earthquakes stand out as a formidable force, arising from the accumulated energy generated at tectonic plate boundaries. When the Earth's layers cannot contain this energy, it is released in the form of seismic waves, resulting in an earthquake.

The magnitude, intensity, and duration of an earthquake play pivotal roles in determining the extent of the destruction and harm it can cause. Between the years 1998 and 2017, earthquakes accounted for nearly 750,000 global fatalities representing over half of all deaths attributed to natural disasters. Additionally, more than 125 million individuals experienced the adverse effects of earthquakes during this period, facing injuries, displacement, homelessness, or evacuation during the emergency phase of the disaster.

There is a growing interest in leveraging the power of Internet of Things to develop cost-effective and scalable earthquake detection systems especially, accelerometers and microcontrollers. Accelerometers, commonly found in smartphones and IoT devices, are sensitive to vibrations and can be used to detect seismic activities when strategically deployed in a network.

One such innovative approach involves integrating accelerometers with Arduino Uno boards, a popular microcontroller platform known for its versatility and ease of programming. By connecting an accelerometer sensor module to an Arduino Uno board, along with additional components like a buzzer for alerts, a breadboard for circuit prototyping, and wires for connections, a simple yet effective earthquake detection system can be constructed.

Arduino's programmability supports for real-time data processing and execution of algorithms to analyze accelerometer readings, distinguishing between usual environmental vibrations and seismic disturbances. When anomalous vibrations indicating potential earthquakes are detected, the system can trigger the buzzer to alert users and initiate early warning protocols. Overall, combining accelerometers with Arduino Uno boards offers a promising solution for developing low-cost, easily deployable earthquake detection systems with the potential for widespread implementation, contributing to improved preparedness and response capabilities in earthquake-prone regions.

Accelerometers, commonly found in smartphones and other electronic devices, measure acceleration forces. By strategically placing these devices in different locations, we can create a distributed sensor network capable of detecting ground movements indicative of seismic activity. This approach not only offers a more accessible means of earthquake monitoring but also has the potential to provide faster detection times compared to traditional methods.

The proposed system aims to design an Earthquake Detector Alarm using Arduino UNO board. This project uses ADXL335 3 axis Accelerometer integrated with Arduino to detect vibrations like tilting, trembling, or any shaking movement of an earthquake. These vibrations are analyzed and check against threshold value. When the reading crosses the threshold value, a warning message is displayed in the LCD display. And at the same time buzzer alarm will be raised and LED glows.

II. RELATED WORK

Authors in [1], assess the effectiveness of a new low-cost triaxial accelerometer prototype utilizing microelectromechanical systems (MEMS) technology for seismicity detection. Two arrays of MEMS sensors were deployed in communication facilities to record nine minor seismic events and one major earthquake. Machine learning techniques have revolutionized earthquake forecasting, offering a pathway to better understand earthquake behavior and refine prediction models [2]. By leveraging large, annotated datasets curated by experts, supervised machine learning models have enhanced seismic monitoring through improved detection, arrival time measurement, phase association, location, and characterization of seismic events. This study underscores the pivotal role of machine learning in advancing earthquake monitoring and prediction capabilities. learning in advancing earthquake monitoring and prediction capabilities.

The study in [3] focuses on developing an intelligent earthquake signal detector to automate traditional disaster response systems. The 3D CNN-RNN architecture captures spatial and temporal information from earthquake signals, enabling the model to differentiate between seismic noise and earthquake signals of varying magnitudes. With an impressive accuracy of 99.057%, sensitivity of 98.488%, and specificity of 99.621%, the model effectively identifies earthquakes and background noise. The authors in [4] discuss the methodology behind Shake Alert, highlighting its phased public rollout, starting with the regional public test in 2018 as Shake Alert 2.0. The system currently provides notifications to over 60 institutional partners across the three mentioned states. By offering advance warning of impending earthquakes, the Shake Alert system aims to enhance preparedness and potentially reduce the impact of disastrous evens on people and infrastructure.

The study in [5] focus on addressing the limitations associated with using smartphones for earthquake detection, primarily due to their high cost and overuse. They propose an alternative solution: a standalone, low-cost earthquake detector that utilizes a carefully chosen low-cost antenna and Wi-Fi connectivity. The researchers in [6] propose a deep learning- based framework aimed at autonomous earthquake detection. This system uses encoded time- frequency representation of the data and successfully retrieves high-order characteristics embedded in three- component seismograms. This innovative approach allows detection of seismic signals across a wide range of local magnitudes accurately.

The study in [7] covers regional and global seismic datasets, prediction logic, and methodologies, revealing inductive logic's prevalence in 62% of studies and deductive logic in 32%. Authors in [8] discusses the development of standalone earthquake warning devices on smartphones. It shows the pointlessness of using smartphone devices only for earthquake detection and talks about problems related to human activities during the day. The authors selected low-power devices, evaluated their performance, and used a simple machine learning technique to train earthquake models.

A deep neural network-based earthquake signal detection system[9] introduced residual structure which combines convolutional layers and bi- directional long-short-term memory(BiLSTM) units. In the study [10] 'the increasing volume of seismic data and the limitations of human analysts' processing capabilities are addressed. Traditional algorithms in time or frequency domains have been used for earthquake interpretation assistance over the past four decades. However, recent advancements involve applying Artificial Neural Networks (ANNs), a machine learning technique, to enhance earthquake detection capabilities. The primary problem addressed in the study [11] is the need for a reliable and efficient earthquake detection system to safeguard lives and property. The proposed solution involves the development of an Arduino-based earthquake detection system utilizing an accelerometer. This system integrates various components such as sensors, a GSM module, and a Wi-Fi module to enhance its functionality and usability. Special attention is paid to the selection of sensors with desirable properties like fast response time and short settling time to improve the system's efficiency in detecting seismic activity.

The study in [12] emphasizes the critical need for accurate earthquake prediction, particularly in obtaining early warnings of seismic events. Despite ongoing research efforts, there exists uncertainty surrounding the feasibility of achieving high forecast accuracy. In response to this challenge, the article suggests exploring the potential of machine learning techniques to enhance short- term earthquake forecasts, drawing inspiration from successful applications in diverse research domains.

A collaborative framework is proposed in [13], integrating IoT sensors for data collection, Edge layer for real-time processing, and magnitude forecasting through the Adaptive Neuro-Fuzzy Inference System (ANFIS) mechanism in cloud layer. Experimental simulations validate the framework's effectiveness, showcasing improvements in classification accuracy, reduced computational delay, increased throughput, reliability, and stability. BLESeis, an innovative and intelligent earthquake detection using IoT sensor [14] proposed. The sensor incorporates a specialized triggering algorithm and an embedded finite impulse response (FIR) low- pass filter to ensure high-fidelity vibration sensing. These features enable accurate capture and processing of vibration data. Subsequently, the acquired vibration signals undergo classification using a bespoke earthquake detection algorithm.

An Arduino-based earthquake detection and warning system that goes beyond seismic monitoring to address secondary hazards like fires and toxic gas leaks [15]. It integrates capacitive three-axis accelerometers for early earthquake detection, IR flame sensors for fire detection, and MQ series air quality sensors for identifying harmful gases. When seismic activity is detected, the system initiates evacuation alerts, activates warning mechanisms for hazardous gas concentrations, and triggers alarms for fires, thus improving overall disaster preparedness. The potential of IoT and cloud infrastructure was explored in[16] in the context of establishing sustainable Earthquake Early Warning Systems (EEWS) at earthquake-prone regions. Beginning with an overview of seismic wave concepts and signal processing fundamentals, it progresses to discuss the IoT-enabled EEWS framework, highlighting IoT networks' role in monitoring EEWS activities and cloud infrastructure for data aggregation, analysis, and alarm dissemination. This study [17] provides a comprehensive review of the mechanisms and operational principles of sensors employed in earthquake monitoring, categorizing them based on the earthquake timeline, sensor mechanisms, and deployment platforms.

Special attention is given to recent sensor platforms such as satellites and UAVs, underscoring their importance in earthquake research and disaster mitigation efforts. Authors in [18], assessed Key performance metrics, including source parameter accuracy, ground-motion forecast precision, and alerting timeliness. They used synthesized earthquake sequences that are generated using real data. Synthesis signals are generated using signals from already recorded earthquakes with varying time shifts. The proposed model analyzes before- and aftershock sequences, simultaneous same kind of events in different regions to comprehensively examine the system's capabilities. They also simulated off-shore and out of network earthquakes in their study. A novel approach to earthquake detection leveraging distributed electromagnetic spectrum detection was proposed in [19]. Traditionally, seismic events are monitored through established stations, but this method proposes utilizing original radio frequency (RF) I/Q sample data and a cumulative distance fluctuation estimation technique for anomaly detection. It addresses the challenge of detecting weak energy electromagnetic radiation emitted due to compression-induced abnormalities prior to earthquakes, a phenomenon associated with the interaction of tectonic plates.

A novel algorithm is designed to detect and locate sources of long-period seismic signals accurately[20]. This algorithm works with a particular focus on data obtained from global seismic networks and stations situated at latitudes beyond 60°. Rayleigh waves at the Earth's surface were analyzed and the model successfully identified over 36,000 events spanning a 13-year period through the application of a delay and stack method. Subsequent meticulous analysis enabled us to differentiate known events from false positives, leading to the identification of more than 1700 previously unknown sources of long-duration of seismic signals. The research outlined in [21] presents a comprehensive framework for assessing infrastructural damages caused post- earthquakes, by integrating remote-sensing data along with advanced machine-learning algorithms. By combining high-resolution building inventory data, maps of earthquake ground shaking intensity, and InSAR images depicting surface- level changes before and after-event, the framework endeavors to classify buildings' damage states in affected regions.

The study [22] underscores the critical importance of implementing an Earthquake Early-Warning System (EEWS) to mitigate the loss of lives during seismic events. Swift assessment of earthquake intensity is vital for effective disaster management and risk mitigation. To expedite this process, the study proposes leveraging an Internet of Things (IoT) network to relay on-site intensity measurements. The proposed approach employs machine learning (ML) techniques, including various linear and nonlinear models, to swiftly determine earthquake intensity within 2 seconds of the P-wave onset. The study in [23] represents a significant advancement in earthquake detection and warning capabilities. Leveraging Machine Learning (ML) algorithms, E3WS swiftly detects, locates, and estimates earthquake magnitudes using just 3 seconds of P-wave data from a single station. Trained on diverse datasets from regions such as Peru, Chile, Japan, and a global dataset, E3WS harnesses the power of six Ensemble ML algorithms operating across temporal, spectral, and cepstral domains.

From above literature survey, it is evident that EEWS is vital and the sensors and various machine learning algorithms are contributing to the detection of earthquakes Maintaining the Integrity of the Specifications.

III. METHODOLOGY

The proposed model for earthquake detection system uses the following components

- ADXL335 Accelerometer
- Arduino Uno Board
- 16 x 2 LCD Display
- Buzzer alarm
- LED

A. ADXL335 Accelerometer

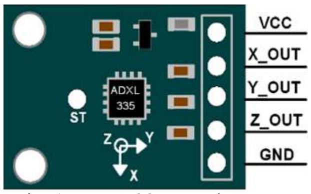

Fig. 1. ADXL335 Accelerometer

Figure 1 shows ADXL335 Accelerometer used in the proposed work. The popular ADXL335 three-axis analog accelerometer integrated circuit, which measures acceleration in X, Y, and Z as analog voltages, is the basis for this accelerometer module. An accelerometer measures the amount of acceleration caused by gravity to determine the angle at which it is tilted with respect to the earth.

B. Unit Arduino Uno Board

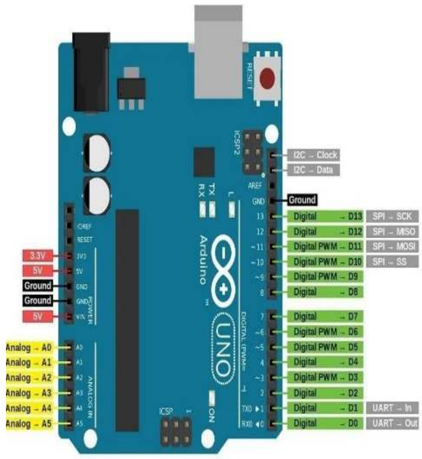

Fig. 2. Arduino UNO Board

The Arduino board as shown in Figure 2 collects the input data and processes to detect earthquake vibrations.

C. 16 x 2 LCD Display

The proposed system uses LCD display as shown in Figure 3 to show the real time feedback and the diagnostic information.

Fig.3. LCD Display

D. Buzzer alarm and LE

Fig 4. Buzzer Alarm

The buzzer alarm as shown in Figure 4, will be used for making alarm when the accelerometer reading crosses the predefined threshold values.

E. Buzzer alarm and LE

Fig 5. LED

When the sensor readings indicate vibrations more than the threshold values, LED glows along with buzzer. Figure 5 depicts LED used in current project.

F. System Architecture

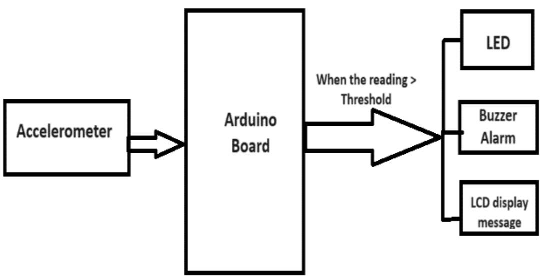

Fig.6. System Architecture

The system architecture is shown in Figure 6. The system is connected and initialized with LCD, buzzer, LED and accelerometer sensor. Multiple samples are collected from accelerometer and average values for x, y, z is calculated. It should Continuously read analog values from the accelerometer sensor. Calculate changes in acceleration for x, y, and z axes. If changes exceed predefined thresholds, it should Activate alerting mechanism and turn on buzzer and LED. And it displays the earthquake alert message on LCD.

We have written code for the Arduino to read analogue values from an accelerometer sensor. We calibrated the sensors to set the zero-gravity level and sensitivity for each axis. We also set a threshold to detect a significant change in speed and programmed the Arduino to trigger an alarm accordingly.

The project IDE on our computer is opened and communicated with the Arduino board. The data from Arduino is read through the communication interface and plot it on the map on the computer screen. The appearance and balance of the image is adjusted based on the accelerometer data and use the real data to see and interact with it. While the image is running, we have to shake the monitor to simulate vibration and watch the image track the intensity and duration of the detected vibration. The model is tested by changing the earthquake intensity and duration of vibration. The threshold and parameter settings are adjusted as needed to improve performance.

G. Arduino code:

Liquid-Crystal library is added to interact with the LCD display and initialize the LCD device with pin connections. Buzzer, LED and accelerometer outputs are defined as variables to store sensor readings, calibration patterns and buzzer activation status. In the setup function, we initialize the LCD, set the communication parameters and display the first message. When the calibration is completed, the LCD screen will be displayed. In the loop function we take the sensor reading for each axis and compare it with the calibration values to calculate the changes in the X, Y and Z axes. These changes are displayed on the LCD screen. If an axis exceeds the specified limit (maxVal or minVal), showing significant vibration, we activate the buzzer and LED. Receive the earthquake alarm by viewing the message on the LCD screen and activate the buzzer for the time (buzTime). The sensor reading is sent to communicate with the processor to prepare the image. Our system constantly monitors speedometer readings and compares them with device readings to detect vibrations. When the vibration exceeds the limit, the earthquake alarm with visual (LED) and auditory (buzzer) signals is activated. Provide instant feedback on the LCD screen and send data into the workflow for further analysis. The basic earthquake test using Arduino is completed, measuring device, LCD screen, buzzer and LED. System sensors measure, monitor vibration in the X, Y and Z axes, generate significant vibration alerts and send information for visualization.

H. Processing code:

Processing and Serial library are imported and defined the variables for the serial port, the horizontal position of the image, and the value of each axis. In the setting function, we set the window size and font size, set the current serial port and start serial communication using the specified port and baud rate. We also set the background color and do not process the input until the "$" sign appears. In the drawing function, we call the serial() function to read data from the serial port. In the Serial() function, we read the incoming data and extract the required value for each axis from the serial

port. We match the extracted results to the image dimensions and draw images using the extracted results.
This code reads the data from the serial port, extracts each axis, and displays the chart using the extracted values.
It will also work on the image and adjust the horizontal position of the image. The Plot function draws a chart using chart values and displays grid lines and labels.

I. Threshold used:

In order to determine the best location for earthquake detection, we recommend to test the system by collecting seismic data recorded in the region where the system is installed. By analyzing data and analyzing the acceleration intervals recorded during the earthquake, we determined the reality to ensure seismic activity and reduce negative impacts. The code can use optional values such as 20 and -20, adjusted according to the characteristics and sensitivity of the accelerometer and the threshold required to detect earthquakes.

J. Connections of the components

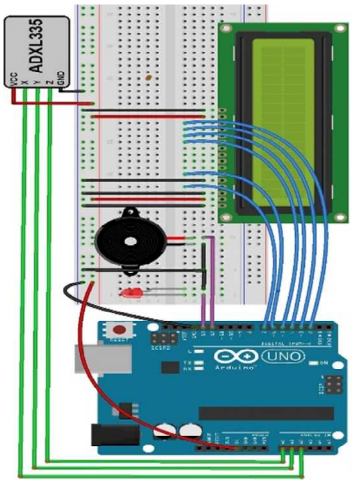

Fig. 8. Circuit diagram

TABLE 1: PIN CONNECTIONS

Component	Pin connection	Purpose
LCD Display	7	Register select pin
LCD Display	6	Enable pin
LCD Display	5	D4 pin
LCD Display	4	D5 Pin
LCD Display	3	D6 Pin
LCD Display	2	D7 Pin
Buzzer	12	Buzzer pin
LED	13	LED pin
Accelerometer	A0	x-Axis is output
Accelerometer	A1	y- Axis is output
Accelerometer	A2	Z-Axis is output

Table 1 shows the pin connections of used in the proposed system as in Figure 8.

IV. Results and Discussions

The earthquake detector system uses an Arduino-based accelerometer (ADXL335) to sense vibrations along the X, Y, and Z axis. When vibrations exceed a threshold on the X-axis, the LED lights up, a buzzer sounds, and an alert message displays on an LCD screen. Pre-earthquake detection is possible by monitoring accelerometer data. The system successfully detects movements and triggers alerts when necessary. The accelerometer's analog outputs are fed into Arduino Uno's ADC pins for processing. In case of significant motion during an earthquake, the system responds with visual (LED), auditory (buzzer), and text alerts. Additionally, the system can visualize earthquake data through graph plotting in software like Processing IDE. The magnitude of the earthquake can be determined using the X, Y and Z values of the accelerometer. Magnitude represents the amount of energy released at the source or location of an earthquake. It is a number created according to the Richter scale and has a single true value no matter where it is measured. The relationship between magnitude and intensity has not yet been established because it depends on factors such as the depth of the earthquake, the composition of the soil around the earthquake, the terrain type and area of the intermediate and measured objects' device or its distance from the epicenter. But it can be derived using the Equation 1.

$$\text{Magnitude} = \log_{10}(\text{sqrt}(X^2 + Y^2 + Z^2)) + 1.5 \qquad (1)$$

Where, X is the reading of x-axis; Y is the reading of y-axis; Z is the reading of z-axis.

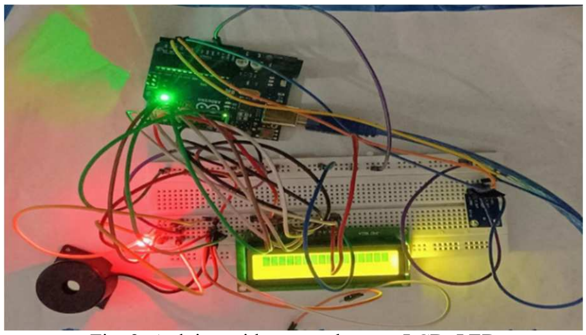

Fig. 9. Arduino with sensor, buzzer, LCD, LED

Fig. 10. Alert message

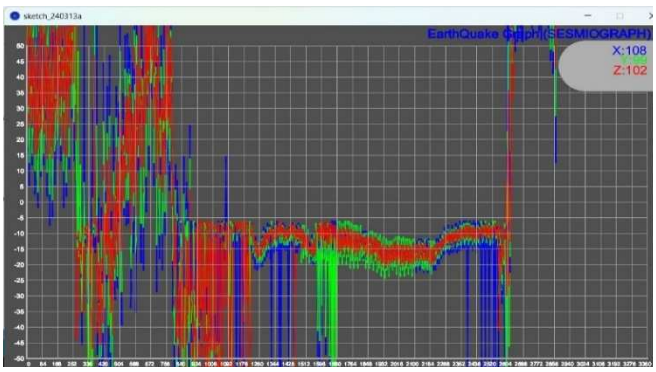

Fig .11. Graph

TABLE II. RECORDED SENSOR READINGS

Time	Date of occurrence	X-axis	Y-axis	Z-axis	Magnitude	Conclusion
12:16AM	04-03-2024	1	2	3	2.07	NORMAL
1:00 PM	04-03-2024	5	6	7	2.52	NORMAL
3:00 PM	04-03-2024	61	68	64	3.55	NORMAL
3:30 PM	04-03-2024	99	110	123	3.78	EARTHQUAKE

Figure 9 shows the proposed model with required components. Figure 10 shows the alert message generated when the Sensor reading value crosses the threshold value used in the system. Figure 11 shows the graph of the values that are recorded. Table II shows the sensor readings at different times.

V. CONCLUSION

The Arduino-based earthquake detection system introduced here aims to mitigate earthquake damage by providing timely alerts to individuals. It is cost-effective, ensuring affordability for a wide range of users. This innovative approach simplifies the automatic detection and classification of seismic activity using a single Arduino-based device. The system offers practical benefits by enhancing preparedness during earthquakes, thereby safeguarding lives and assets. Its low power consumption makes it suitable for household applications. Moreover, the system's adaptability allows for alternative uses such as knock-and-shake detection in ATMs, vehicles, or as door-break alarms. Additionally, the use of ultra-compact, high-accuracy earthquake detection sensor modules enhances the system's ability to detect vibrations accurately during seismic events.

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